



**COMSATS University Islamabad,
Abbottabad Campus**

SOFTWARE REQUIREMENTS SPECIFICATION

(SRS DOCUMENT)

For

Depression Alert AI

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Revision History

Name	Date	Reason for Changes	Version
Muhammad Salahudin Khan Muhammad Azan			

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1. Introduction

1.1 Purpose

This document specifies the software requirements for DepressionAlert AI, a web-based system that analyzes user-provided social media text to identify early signs of depression. It defines the functional and non-functional requirements, use cases, and interface requirements that will guide the design, development, and testing of the system.

1.2 Scope

DepressionAlert AI allows registered users to submit social media posts, either by pasting text directly or uploading a CSV file, for analysis using Natural Language Processing and Machine Learning. The system extracts emotional and linguistic features from the submitted text, computes a depression risk score, and raises alerts when the score exceeds a defined threshold. Results, trends, and alerts are presented through a web dashboard.

The system provides supportive monitoring and awareness only. It does not perform medical diagnosis or treatment and all clinical decisions remain the responsibility of qualified mental health professionals.

2. Overall Description

2.1 Product Perspective

DepressionAlert AI is a new, standalone web application. It is not a replacement for clinical assessment tools; rather, it complements traditional mental health evaluation by providing continuous, text-based monitoring between clinical visits.

2.2 Operating Environment

OE-1: The system shall operate correctly on current versions of Google Chrome, Mozilla Firefox, and Microsoft Edge on desktop and mobile browsers.

OE-2: The backend API and machine learning services shall run on a Linux-based server environment with Python 3.11 and Node.js 20.x LTS.

2.3 Design and Implementation Constraints

CO-1: The frontend shall be implemented using React 18.x.

CO-2: The backend API shall be implemented using Node.js, with AI, NLP, and Machine Learning components implemented in Python 3.11 using scikit-learn and spaCy/NLTK.

CO-3: The machine learning model shall be trained using publicly available, ethically sourced mental health text datasets.

3. Requirement Identifying Technique

Use case analysis was selected as the primary requirement identifying technique for DepressionAlert AI, as it is well suited to an interactive, end-user-facing web application.

3.1 Use Case Diagram

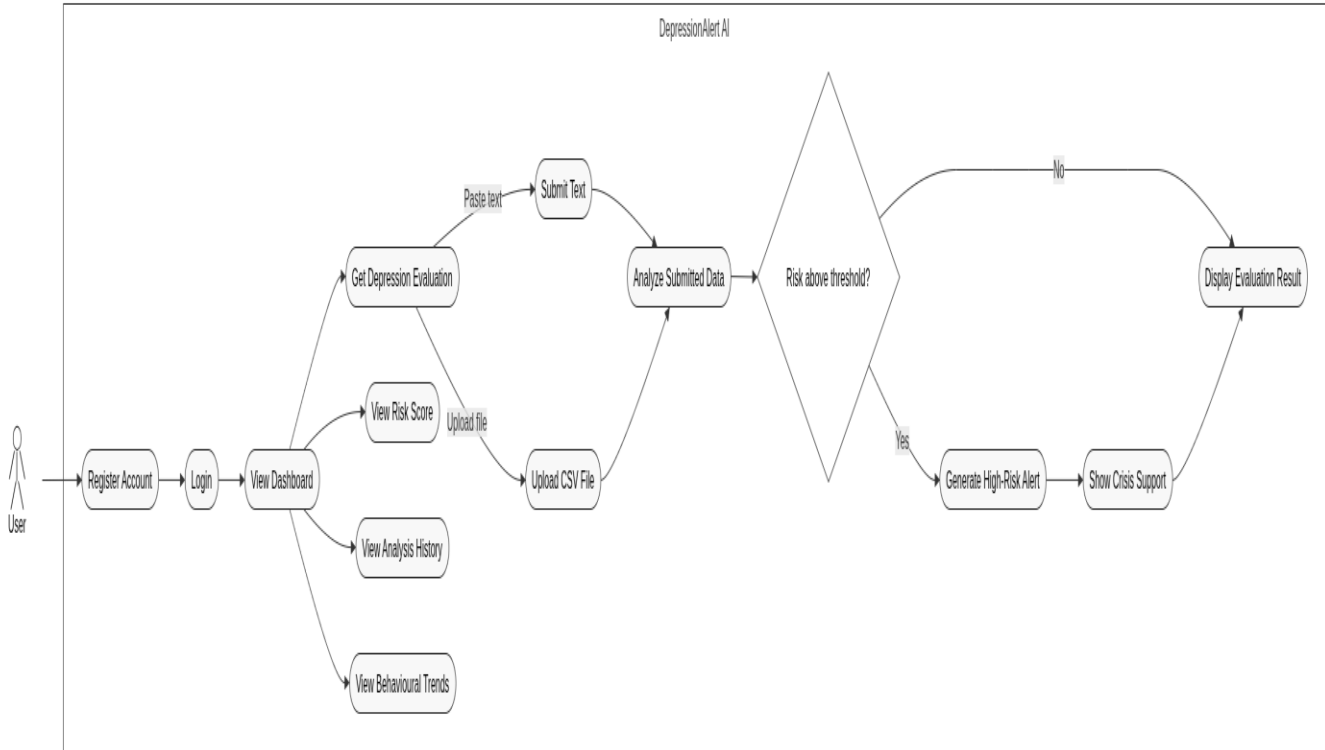


Figure: Use Case diagram

3.2 Use Case Description

UC-1: Register Account

Actor(s)	New User
Description	A new user creates an account to access the DepressionAlert AI platform and its depression-evaluation features.
Trigger	User selects Register Account .
Preconditions	PRE-1. User does not already have an account. PRE-2. User has access to a valid email address.
Postconditions	POST-1. Submitted text is stored against the user's profile. POST-2. Text is queued for NLP preprocessing.
Normal Flow	1. User opens the registration screen. 2. User enters their name, email, and password. 3. System validates the entered information. 4. System creates the account. 5. System displays a registration confirmation and redirects the user to Login.
Exceptions	Email is already registered: system displays an error message. Required information is missing: system asks the user to complete all required fields. Password does not meet requirements: system asks the user to enter a valid password.

UC-2: Login

Actor(s)	Registered User
Description	A registered user enters their credentials to securely access the system.
Trigger	User selects Login after opening the application or completing registration.
Preconditions	PRE-1. User is authenticated. PRE-2. At least one analysis has been completed.
Postconditions	PRE-1. User has a registered account. PRE-2. User is not currently logged in.
Normal Flow	1. User opens the Login screen. 2. Users enter their email and password. 3. System validates the credentials. 4. System creates an authenticated session. 5. System redirects the user to the Dashboard.
Exceptions	Credentials are incorrect: system displays an error message. Account does not exist: system prompts the user to register. Authentication service is unavailable: system asks the user to try again later.

UC-3: View Dashboard

Actor(s)	Registered User
Description	A user views the main dashboard and selects from the available depression-monitoring features.
Trigger	User successfully logs in or selects Dashboard from the navigation menu.
Preconditions	PRE-1. User is authenticated.
Postconditions	POST-1. Dashboard information is displayed. POST-2. User can select an available system feature.
Normal Flow	1. User opens the Dashboard. 2. System retrieves the latest available user information. 3. System displays the available options. 4. User selects one of the following: Get Depression Evaluation, View Risk Score, View Analysis History, or View Behavioural Trends.
Exceptions	Dashboard data cannot be loaded: system displays an error and provides a retry option. No analysis exists: system prompts the user to complete a depression evaluation.

UC-4: Get Depression Evaluation

Actor(s)	Registered User
Description	A user submits social media text manually or uploads a CSV file so the system can evaluate it for depression-related indicators.
Trigger	User selects Get Depression Evaluation from the Dashboard.
Preconditions	PRE-1. User is authenticated. PRE-2. User has accepted the data-processing consent. PRE-3. The submitted text or CSV file contains user-authorized data.
Postconditions	POST-1. Valid text is stored against the user's account. POST-2. The text is processed using NLP. POST-3. A risk score and risk level are generated. POST-4. The result is stored in the analysis history.
Normal Flow	1. User opens the Depression Evaluation screen. 2. User selects Submit Text or Upload CSV File . 3. User enters text or selects a CSV file. 4. System checks for empty, malformed, invalid, or duplicate records. 5. System cleans and processes the valid text using NLP. 6. System extracts sentiment and relevant text features. 7. Depression Detection Model calculates the risk score. 8. System stores and displays the evaluation result.
Exceptions	Submitted text is empty: system asks the user to enter text. Uploaded file is not a valid CSV: system rejects the file and displays an error. Duplicate records are detected: system removes or flags duplicate records. Analysis fails: system displays an error and allows the user to retry.

UC-5: View Risk Score

Actor(s)	Registered User
Description	A user views the latest depression risk score and its corresponding risk level.
Trigger	User selects View Risk Score from the Dashboard.
Preconditions	PRE-1. User is authenticated. PRE-2. At least one depression evaluation has been completed.
Postconditions	POST-1. The latest risk score and risk level are displayed. POST-2. The stored analysis result remains unchanged.
Normal Flow	1. User selects View Risk Score. 2. System retrieves the latest analysis result. 3. System displays the risk score. 4. System displays the risk level as Low, Medium, or High. 5. System displays a brief explanation of the result.
Exceptions	5.0.E1 No analysis result exists: system prompts the user to complete an evaluation. 5.0.E2 Result cannot be retrieved: system displays an error message.

UC-6: View Analysis History

Actor(s)	Registered User
Description	A user views the records of their previous depression evaluations and risk results.
Trigger	User selects View Analysis History from the Dashboard.
Preconditions	PRE-1. User is authenticated. PRE-2. At least one analysis result exists.
Postconditions	POST-1. Previous analysis records are displayed. POST-2. No stored record is modified.
Normal Flow	1. User opens Analysis History. 2. System retrieves the user's previous analysis results. 3. System arranges the records by analysis date. 4. System displays the date, risk score, risk level, and sentiment result. 5. User may select a record to view its details.
Exceptions	6.0.E1 No previous analysis exists: system displays an empty-state message. 6.0.E2 History cannot be retrieved: system displays an error and retry option.

UC-7: View Behavioral Trends

Actor(s)	Registered User
Description	A user views changes in their risk scores, sentiment, and emotional patterns across previous analyses.
Trigger	User selects View Behavioral Trends from the Dashboard.
Preconditions	PRE-1. User is authenticated. PRE-2. Sufficient previous analysis results are available for comparison.
Postconditions	POST-1. Behavioral trends are displayed. POST-2. Existing analysis records remain unchanged.
Normal Flow	1. User opens Behavioral Trends. 2. System retrieves the user's previous analysis results. 3. Behavioral Analyzer compares current and previous results. 4. System identifies changes in risk score, sentiment, and emotional indicators. 5. System displays the trends using charts and summaries.
Exceptions	7.0.E1 Insufficient analysis records exist: system asks the user to complete more evaluations. 7.0.E2 Trend analysis fails: system displays an error message.

UC-8: Receive High-Risk Alert and Crisis Support

Actor(s)	Registered User
Description	The system generates a warning and displays crisis-support resources when the calculated risk score exceeds the configured threshold
Trigger	Depression Detection Model returns a risk score above the configured high-risk threshold.
Preconditions	PRE-1. A depression evaluation has been completed. PRE-2. A valid risk score has been generated. PRE-3. The risk score exceeds the configured threshold.
Postconditions	POST-1. A high-risk alert is created and stored. POST-2. The alert is displayed to the user. POST-3. Crisis-support resources are displayed. POST-4. The alert appears in the user's dashboard and history.
Normal Flow	1. Depression Detection Model generates the risk score. 2. System compares the score with the configured threshold. 3. System identifies the result as high risk. 4. System creates and stores a high-risk alert. 5. System displays the alert to the user. 6. System displays crisis helplines, support resources, and recommended immediate actions. 7. User views or dismisses the alert.
Exceptions	Alert cannot be stored: system logs the failure and retries. Support resources cannot be loaded: system displays the available emergency message and retry option. Alert display fails: system keeps the alert active for the next dashboard visit.

4. Specific Requirements

This section itemizes the functional requirements of DepressionAlert AI, organized by module.

FR-1: User Registration

Identifier	FR-1
Title	User Registration
Requirement	The system shall allow a new user to create an account using their name, email, and password.
Rationale	Enables users to securely access the platform and associate analysis results with their accounts.

FR-2: User Login

Identifier	FR-2
Title	User login
Requirement	The system shall authenticate registered users using valid login credentials before allowing access to protected features.
Rationale	Protects user information and analysis records from unauthorized access.

FR-3: Consent Verification

Identifier	FR-3
Title	Consent Verification
Requirement	The system shall verify that the user has granted data-processing consent before accepting or processing submitted text.
Rationale	Supports ethical, permission-based, and privacy-conscious data handling.

FR-4: Manual Text Submission

Identifier	FR-4
Title	Manual Text Submission
Requirement	The user shall be able to paste social media post text directly into the platform for analysis.
Rationale	Allows analysis without requiring users to export their social media data.

FR-5: CSV Upload

Identifier	FR-5
Title	CSV Upload
Requirement	The user shall be able to upload a CSV file containing multiple posts for batch analysis.
Rationale	Supports analysis of larger volumes of historical posts.

FR-6: Submission Validation

Identifier	FR-6
Title	Submission Validation
Requirement	The system shall validate submitted text and CSV records to identify empty, malformed, invalid, or duplicate entries before processing.
Rationale	Prevents unsuitable or repeated data from affecting analysis results.

FR-7: Text Preprocessing

Identifier	FR-7
Title	Text Preprocessing
Requirement	The system shall clean valid text by removing unnecessary symbols, stop words, and irrelevant characters before analysis.
Rationale	Produces clean and meaningful text for feature extraction and model analysis.

FR-8: Sentiment and Feature Extraction

Identifier	FR-8
Title	Sentiment and Feature Extraction
Requirement	The system shall tokenize processed text and extract sentiment, emotional, and linguistic features from each valid submission.
Rationale	Produces the feature set required by the depression-detection model.

FR-9: Risk Score Generation

Identifier	FR-9
Title	Risk Score Generation
Requirement	The system shall use a trained machine-learning model to generate a depression-related risk score and risk level for each completed analysis.
Rationale	Provides the main analysis result for identifying possible depression-related indicators.

FR-10: Behavioral Trend Analysis

Identifier	FR-10
Title	Behavioral Trend Analysis
Requirement	The system shall compare a user's current and previous analysis results to identify changes in sentiment, risk scores, and emotional patterns over time.
Rationale	Helps users understand changes in their emotional and linguistic patterns.

FR-11: Threshold-Based Alerting

Identifier	FR-11
Title	Threshold-Based Alerting
Requirement	The system shall automatically generate and store a high-risk alert when an analysis result exceeds the configured risk threshold.
Rationale	Provides timely awareness when strong depression-related patterns are detected.

FR-12: Crisis Support Resources

Identifier	FR-12
Title	Crisis Support Resources
Requirement	The system shall display appropriate crisis-support information and supportive resources whenever a high-risk alert is generated.
Rationale	Provides immediate access to supportive information when high-risk indicators are detected.

FR-13: Dashboard Display

Identifier	FR-13
Title	Dashboard Display
Requirement	The system shall display the user's latest risk score, current risk level, recent alert status, and available analysis options on the dashboard.
Rationale	Provides a clear summary of the user's current monitoring information.

FR-14: Analysis History Display

Identifier	FR-14
Title	Analysis History Display
Requirement	The system shall allow the user to view previous analysis results, including analysis date, risk score, risk level, and sentiment result.
Rationale	Allows users to review previous results and monitor changes over time.

5. Quality Attributes

5.1 Usability

USE-1: Simple Text Submission

The system shall allow an authenticated user to access the Depression Evaluation screen and submit text for analysis in no more than three interactions from the Dashboard.

USE-2: Clear Dashboard Information

The Dashboard shall display the user's latest risk score, risk level, recent alert status, and behavioural trend summary without requiring navigation to another page.

USE-3: Input Guidance and Validation

The system shall provide clear labels, instructions, validation messages, and error feedback for manual text submission and CSV file upload.

USE-4: Responsive Interface

The user interface shall remain usable and readable on desktop computers, laptops, tablets, and smartphones using modern web browsers.

USE-5: Consistent Navigation

The system shall provide consistent navigation, button placement, terminology, colours, and interface elements across all screens.

5.2 Performance

PER-1: Text Analysis Response Time

The system shall generate and display a risk score for a valid single-text submission within five seconds under normal operating conditions.

PER-2: Dashboard Loading Time

At least 95% of Dashboard requests shall load completely within three seconds when accessed through an internet connection of 10 Mbps or faster under normal load.

PER-3: CSV Processing Feedback

The system shall acknowledge a valid CSV upload within three seconds and display its processing status until the batch analysis is completed.

PER-4: Concurrent User Support

The system shall support multiple concurrent users without causing significant degradation in response time under the expected operational load.

5.3 Security and Privacy**SEC-1: Secure Communication**

All user credentials, submitted text, uploaded files, analysis results, and alerts shall be encrypted during transmission using HTTPS with TLS 1.2 or higher.

SEC-2: Consent-Based Processing

The system shall accept and process submitted text only after the user has explicitly granted data-processing consent. The user shall be able to revoke this consent.

SEC-3: Password Protection

User passwords shall be stored using a secure one-way password-hashing algorithm and shall never be stored or transmitted as plain text.

SEC-4: Authentication and Authorization

The system shall require authentication before allowing users to submit text, upload CSV files, or access analysis results, alerts, behavioural trends, and history.

SEC-5: User Data Isolation

The system shall ensure that users can access only their own profile, submissions, analysis results, alerts, and history.

SEC-6: Secure File Validation

The system shall validate uploaded CSV files for permitted format, file size, malformed content, and potentially unsafe data before processing.

SEC-7: Session Security

The system shall securely terminate the user session after logout and shall prevent unauthorized access using expired or invalid authentication tokens.

6. External Interface Requirements

6.1 User Interfaces

The system shall provide a responsive web-based user interface accessible through modern desktop and mobile web browsers. The interface shall follow a consistent design in terms of colours, typography, navigation, buttons, forms, and layout.

The system shall include the following main interfaces:

- **Registration and Login Interface:** Allows users to create an account and securely log in to the system.
- **Dashboard Interface:** Displays the latest depression risk score, risk level, recent alerts, analysis history, and behavioural trends.
- **Daily Log / Depression Evaluation Interface:** Allows users to manually paste social media text or upload a CSV file for analysis.

- **Analysis Result Interface:** Displays the generated risk score, risk level, sentiment result, and identified emotional or linguistic indicators.
- **Sentiment Alert Interface:** Displays a high-risk warning when the calculated risk score exceeds the configured threshold.
- **Crisis Support Interface:** Provides crisis-support information, emergency contacts, calming exercises, and educational resources.
- **Analysis History Interface:** Allows users to view their previous analysis results.
- **Behavioural Trends Interface:** Displays changes in risk scores, sentiment, and emotional patterns over time.
- **Privacy and Profile Interface:** Allows users to review privacy information and manage their account details.

6.2 Software Interfaces

The frontend shall be developed using React.js and shall communicate with the Node.js backend through a RESTful API using JSON-formatted requests and responses.

The backend shall provide interfaces for:

- User registration and authentication
- Manual text submission
- CSV file upload
- Submission validation
- Analysis-result retrieval
- Analysis-history retrieval
- Behavioral-trend retrieval
- Alert generation and retrieval
- Crisis-support resource retrieval

The Node.js backend shall communicate with the internal Python-based NLP and machine-learning service through a protected service interface.

The Python service shall perform:

- Text cleaning and preprocessing
- Tokenization
- Sentiment analysis
- Emotional and linguistic feature extraction
- Behavioral-pattern analysis
- Depression-risk score generation

- Risk-level classification

The backend shall store user accounts, text submissions, analysis results, behavioural trends, and alerts in the system database.

6.3 Hardware Interfaces

The system shall not require any dedicated hardware devices.

Users shall access the system through standard client devices, including:

- Desktop computers
- Laptop computers
- Tablets
- Smartphones

Each client device shall require:

- A modern web browser
- An active internet connection

The server environment shall provide sufficient processing, memory, and storage resources to run the frontend, backend, database, NLP components, and trained machine-learning model.

6.4 Communications Interfaces

The system shall use HTTPS for all communication between the client application and the backend server.

The frontend and backend shall exchange data through RESTful API requests and responses using JSON.

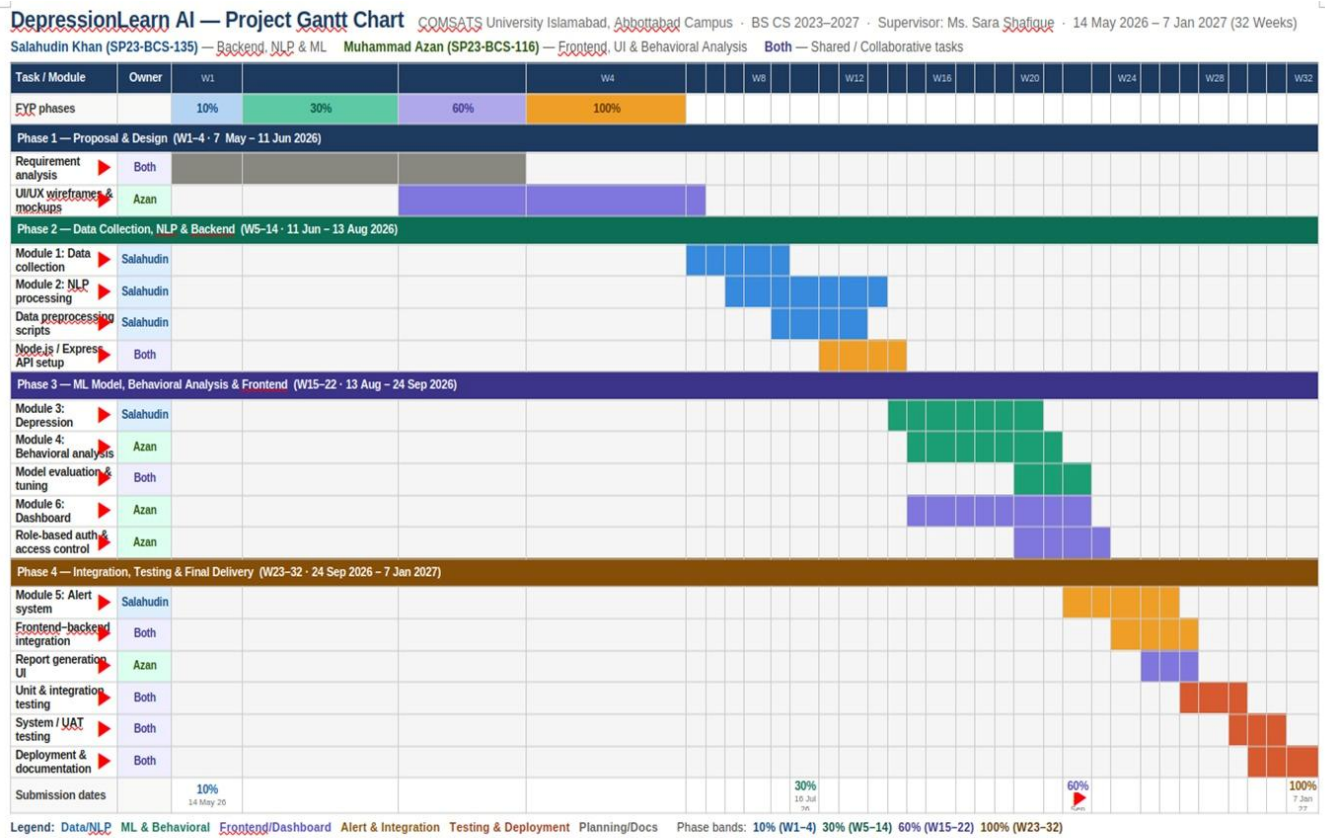
CSV files shall be transmitted through secure multipart file-upload requests.

Communication between the Node.js backend and the Python-based NLP and machine-learning service shall occur through a protected internal API.

High-risk alerts shall be delivered through in-application notifications and displayed on the user dashboard.

All communication containing user information, submitted text, analysis results, or alerts shall be transmitted through encrypted connections.

7. Project Gantt Chart



8. References

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